

Chiao-Tung (Carol) Chan

Staff Software Engineer / R&D Manager, Autonomous Systems

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PROFESSIONAL SUMMARY

ISO 26262 Certified (AFSP) R&D Manager and Autonomous Systems Architect with 7+ years of full-stack experience spanning algorithm design (Localization & Tracking) to low-level hardware optimization (Edge AI & Compute). Proven track record of leading cross-functional teams to build 0-to-1 autonomous driving systems, deploying robust Visual-Inertial Localization and Sensor Fusion pipelines in international testing fields. Deep expertise in C++ system architecture, mitigating complex HW/SW bottlenecks (zero-copy memory management, TensorRT acceleration), and establishing scalable, safety-critical CI/CD infrastructure.

CERTIFICATIONS & PATENTS

- **Automotive Functional Safety Professional (AFSP) – ISO 26262**, SGS TÜV Saar — Valid through Oct 2025.
- **3D Object Detection Method and System** (Sole Inventor), Dec 2024 — Taiwan App. 113150958 / Japan App. 2025-069543 (Pending).
- **Camera Calibration Method Based on Vehicle Localization** (Lead Inventor), Dec 2023 — US Patent 12,400,367; Taiwan I882582; Japan 2024-071947 (Allowed).
- **Safe Following Distance Estimation System and Method** (Co-Inventor), Dec 2021 — US Patent 11,891,063.

EXPERIENCE

Industrial Technology Research Institute (ITRI)

R&D Manager, Self-Driving Vehicle Department

Hsinchu, Taiwan

Apr 2019 – Jun 2026

1. Algorithm Development: Localization & Tracking

- Spearheaded the 0-to-1 architecture and cross-border deployment of a safety-critical Visual-Inertial Localization pipeline (Camera/IMU/Odometry), establishing RTK-backed ground-truth validation infrastructure. Formulated production-grade state-estimation strategies under severe edge-device compute constraints, driving international commercialization at the Toyota Driving School (Japan) and Taiwan testing fields within a tight 12-month window.
- Defined the enterprise technical vision for a modular C++ Object-Tracking Framework. Architected extensible abstract interfaces for heterogeneous sensor modalities (Radar x/y/z/rv, Camera IDs) using IMM-UKF algorithms, creating a 2x development-velocity multiplier that unblocked parallel delivery for both Localization and Fusion engineering teams.
- Pioneered a novel L-Shape tracking methodology leveraging Iterative Closest Point (ICP) to directly infer precise vehicle yaw, bypassing legacy non-linear state-estimator limitations in degenerate scenarios. Translated this algorithmic breakthrough into a commercially high-value, sole-inventor 3D Object Detection patent.

2. AI Acceleration & Edge Compute Optimization

- Overhauled the deep-learning inference architecture by transitioning legacy data structures to a modern Reactive (Rx) streaming paradigm. Engineered a custom zero-copy memory-management framework leveraging TensorRT, maximizing sustained GPU utilization to 95% while strictly guaranteeing deterministic, safety-critical real-time latency limits.
- Championed a critical HW/SW co-design initiative to resolve catastrophic system-level thermal throttling and OOM failures. Defined rigorous end-to-end latency budgets (e.g., 28 Hz for visual localization) and designed adaptive algorithmic filters to dynamically prune stale sensor data, directly steering the physical thermal redesign of the vehicle compute trunk.
- Orchestrated a strategic edge-compute offloading roadmap, leading technical integration with external hardware partners (Chimei). Architected a decentralized computing paradigm by migrating image compression and neural-network inference directly to edge ASICs, eliminating bulky frame grabbers and drastically alleviating central compute bottlenecks.

3. Technical Leadership & Infrastructure (DevOps)

- Directed technical execution for a cross-functional engineering team, scaling the unit from 4 to 10+ members. Formulated a comprehensive 4-year technical roadmap focused on resilient system scalability and continuous automated functional-safety validation.
- Orchestrated a 6-month ROS1 to ROS2 migration for a junior team — defined strict SOPs and decomposed tasks (message interface, TF, bag conversion), and mandated automated CI/CD pipelines on AWS and internal servers for bag-file playback, eradicating regression bugs during deployment.
- Engineered an automated Active-Learning data pipeline that runs offline bag-file playbacks against ML object detectors. Designed custom mismatch-trigger logic (Track vs. Future-Track discrepancies) to programmatically mine hard-negative edge cases, establishing a closed-loop data engine for continuous model refinement.

EDUCATION

National Yang Ming Chiao Tung University (NYCU / NCTU)

M.S. in Electrical Engineering

Hsinchu, Taiwan

Sep 2016 – Jan 2019

National Yang Ming Chiao Tung University (NYCU / NCTU)

B.S. in Electrical Engineering

Hsinchu, Taiwan

Sep 2012 – Jun 2016

PUBLICATIONS

- **RMSeg-UDA: Unsupervised Domain Adaptation for Road Marking Segmentation**, ICRA 2025. Y.-C. Cai, H.-C. Hsiao, W.-C. Chiu, H.-Y. Lin, **C.-T. Chan**.
- **FuseRoad: Enhancing Lane Shape Prediction via Semantic Knowledge Integration**, IV 2025. H.-C. Hsiao, Y.-C. Cai, H.-Y. Lin, W.-C. Chiu, **C.-T. Chan**, C.-C. Wang.
- **MemPromptTSS: Persistent Prompt Memory for Iterative Multi-Granularity Time Series State Segmentation**, arXiv 2025. C. Chang, M.-C. Lo, **C.-T. Chan**, W.-C. Peng, T.-F. Chen.
- **TimeDRL: Disentangled Representation Learning for Multivariate Time-Series**, ICDE 2024. C. Chang, **C.-T. Chan**, W.-Y. Wang, W.-C. Peng, T.-F. Chen.
- **Self-Supervised Learning of Disentangled Representations for Multivariate Time-Series**, NeurIPS 2024 Workshop (Self-Supervised Learning — Theory and Practice). C. Chang, **C.-T. Chan**, W.-Y. Wang, W.-C. Peng, T.-F. Chen.
- **RLMD: A Dataset for Road Marking Segmentation**, ICCE-Taiwan 2023. H.-C. Hsiao, Y.-C. Cai, H.-Y. Lin, W.-C. Chiu, **C.-T. Chan**.

CORE COMPETENCIES

- **Algorithm & Autonomy:** Visual-Inertial Localization, Sensor Fusion, Object Tracking (IMM-UKF, ICP)
- **AI Acceleration & Compute:** CUDA, TensorRT, Memory Management (Zero-copy), HW/SW Co-design
- **Infrastructure & MLOps:** C++, Python, ROS1/ROS2, AWS CI/CD, Nix, Active-Learning Data Pipelines
- **Leadership & Compliance:** ISO 26262 Functional Safety, Team Scaling, Roadmap Planning, Agile/SOP